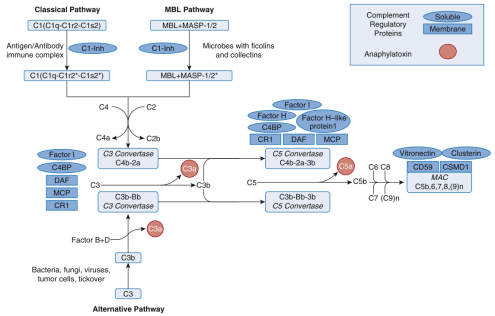

Fig. 36.1 The complement system and its alternative pathway. The alternative pathway of complement (AP) begins with spontaneous hydrolysis of C3 and the production of C3b. C3b combines with Bb to form the C3 convertase. The C3 convertase may go on to form the C5 convertase (C3bBbC3b) after the addition of another C3b or it may feedback to make additional C3b in an amplification loop. C5 convertase will cleave C5, producing C5b, the first protein in the terminal complement complex. The ultimate products of the AP include two important anaphylatoxins (C3a and C5a) and the C5b-9 complex. Factor H is the major negative regulator of the AP, and dysregulation of FH is the most common genetic cause of aHUS. (Created in BioRender. Licht C. 2025. https://BioRender.com/0w57eov.)

Fig_36_1

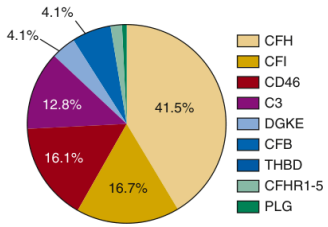

Fig. 36.2 Yield of genetic testing for a large cohort of patients with clinical features of atypical hemolytic uremic syndrome (aHUS). The data presented are from Osborne et al,[22] which collated genetic testing results from 3108 patients from 6 centers. The top panel illustrates the percentage of positive tests for given aHUS genes when compared with all positive tests. This data are also displayed graphically in the pie chart in the bottom panel.

Fig_36_2

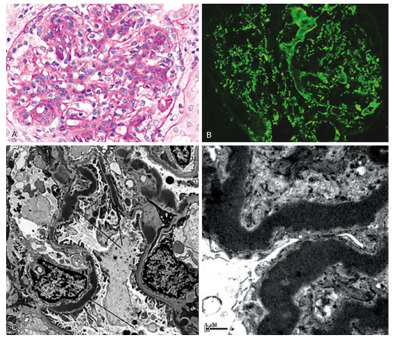

Fig. 36.3 Dense deposit disease. (A) Diffuse mesangial hypercellularity with segmental endocapillary hypercellularity and pink ribbon-like enhancement of glomerular basement membranes (periodic acid-Schiff stain, original magnification 600×); (B) C3 granular deposits diffusely in mesangial areas and segmentally along glomerular basement membranes (fluorescein-conjugated antihuman C3, original magnification 600×); (C) segmental electron dense transformation of glomerular basement membranes and mesangial areas (arrows, unstained grid, and original magnification 4000×); and (D) diffuse electron dense transformation of glomerular basement membranes (unstained grid, original magnification 12,000×).

Fig_36_3

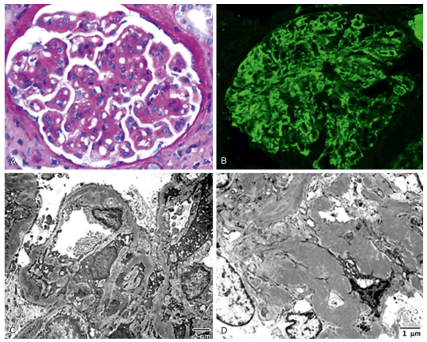

Fig. 36.4 C3 glomerulonephritis. (A) diffuse mesangial hypercellularity with endocapillary hypercellularity and thick glomerular basement membranes (periodic acid- Schiff stain, original magnification 600×); (B) C3 granular deposits diffusely in mesangial areas and along glomerular basement membranes (fluorescein-conjugated antihuman C3, original magnification 600×); (C) segmental charcoal gray transformation of glomerular basement membranes and mesangial areas (unstained grid, original magnification 4000×); and (D) diffuse charcoal gray transformation predominantly in mesangial areas but also within glomerular basement membranes (unstained grid, original magnification 8000×).

Fig_36_4

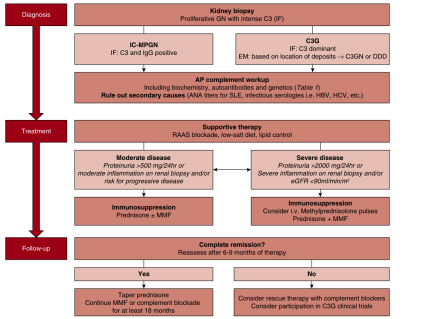

Fig. 36.5 Algorithm for Diagnosis, treatment and follow-up in IC-MPGN and C3G patients. Complement blockers targeting CFB (C3 convertase) and C3 are now available for the treatment of C3G and/or IC-MGPN patients >12 years including patients with post-transplant disease recurrence. Treatment algorithms or protocols–other than the ones used in the clinical trials–are not available, yet. C3 and CFB inhibitors may be used for primary treatment (with or without prednisone/MMF), for treatment escalation in patients not responding to prednisone/MMF, or in patients with severe or progressive disease. While drugs in transplant patients are currently only approved for disease recurrence, preemptive treatment aiming to prevent recurrence should be considered. Also, there are no guidelines defining treatment duration. As for now, therapy duration should be decided on an individual basis and can be discussed with experts.

Fig_36_5

Table 36.1 Overview of Complement Amplifying Conditions that can Associate With Thrombotic Microangiopathy (TMA)

Pregnancy-associated TMA (pTMA)
Post–hematopoietic stem cell transplant–associated TMA or transplant TMA (TA-TMA)
TMA post solid organ transplantation
TMA associated with metabolic disease: cobalamin C deficiency
TMA associated with systemic autoimmune diseases
- Systemic lupus erythematosus
- Antineutrophil cytoplasmic antibody–associated vasculitis
- Antiphospholipid syndrome
- Systemic sclerosis
- Mixed connective tissue disorder
- Sjögren syndrome
- Inflammatory bowel disease

TMA associated with drugs
- Quinine
- Anti-VEGF inhibitor
- Chemotherapy such as gemcitabine, mitomycin, and cisplatin
- Calcineurin inhibitors
- Gene therapy

TMA associated with malignant hypertension
TMA associated with infections
- Bacteria associated with a gastrointestinal infection, such as Shiga toxin producing *Escherichia coli*, *Shigella dysenteriae*, *Campylobacter jejuni*, *Moraxella osloensis*
- Viruses, such as cytomegalovirus, Epstein-Barr virus, varicella zoster virus, parvovirus B19, human immunodeficiency virus, influenza
- Bacteria causing respiratory tract infections, such as Bordetella pertussis, Streptococcus pneumoniae, Mycoplasma pneumoniae, COVID-19
- Fungal infections in immunocompromised hosts

TMA associated with other renal diseases
- C3G/MPGN
- IgA Nephropathy

Table_36_1

Table 36.2 Eculizumab Dosing Regimen (Induction and Maintenance)

Body Weight	Induction	Maintenance
40 kg and above	900 mg weekly for 4 weeks	1200 mg at week 5; then 1200 mg every 2 weeks
30 to <40 kg	600 mg weekly for 2 weeks	900 mg at week 3; then 900 mg every 2 weeks
20 to <30 kg	600 mg weekly for 2 weeks	600 mg at week 3; then 600 mg every 2 weeks
10 to <20 kg	600 mg week 1	one dose 300 mg at week 2; then 300 mg every 2 weeks
5 to <10 kg	300 mg week 1	one dose 300 mg at week 2; then 300 mg every 3 weeks

Table_36_2

Table 36.3 Secondary Causes of IC-MPGN

Immunoglobulin mediated

Infectious diseases
- Hepatitis B, hepatitis C, hepatitis E, Epstein Barr virus, and human immunodeficiency virus
- Endocarditis, shunt nephritis, visceral abscess, and empyema
- Tuberculosis, leprosy
- Brucellosis, Q fever
- Malaria, schistosomiasis
- Onchocerciasis; filariasis, *Loa loa* infections

Systemic immune diseases
- Cryoglobulinemia
- Systemic lupus erythematosus
- IgA vasculitis
- Sjögren syndrome (secondary to cryoglobulinemia)
- Hypocomplementemic urticarial vasculitis
- Rheumatoid arthritis
- X-linked agammaglobulinemia
- Neoplasms/dysproteinemias
- Plasma cell dyscrasia
- Monoclonal gammopathy of unknown significance
- Fibrillary glomerulopathy
- Immunotactoid glomerulonephritis
- Proliferative glomerulonephritis with monoclonal immunoglobulin deposition
- Proliferative glomerulonephritis with isolated light chain deposition
- MPGN with masked monotypic immunoglobulin deposits
- Light-chain and/or heavy-chain deposition disease
- Leukemias and lymphomas (with or without cryoglobulinemia)
- Bone marrow transplant
- Carcinomas, Wilms tumor, and malignant melanoma
- Angiofollicular lymph node hyperplasia with or without TAFRO syndrome
- Sinus histiocytosis with massive lymphadenopathy

Other
- α-1-antitrypsin deficiency with severe liver disease
- Gluten-sensitive enteropathy (celiac disease)
- Immunoglobulin negative
- Thrombotic microangiopathy including drug-induced forms
- Sickle-cell disease
- Transplant glomerulopathy
- POEMS syndrome

Table_36_3

Table 36.4 Autoantibodies Associated With C3G/IC-MPGN

Autoantibodies	Incidence	Coexisting With C3Nef	Effect on Complement	Routine Testing
C3NeF	Common		Stabilizes AP C3 convertase	Yes
C4NeF	Rare	Yes	Stabilizes CP C3 and AP and CP C5 convertases	No
C5NeF	Rare	Yes	Stabilizes AP C5 convertase	No
Anti-factor B	Rare	No	Stabilizes AP C3 convertase	No
Anti-C3b	Rare	No	Stabilizes AP C3 convertase	No
Anti-factor H	Rare	Yes	Fluid phase AP regulation	Yes

Table_36_4

Table 36.5 Pathogenic Gene Variants, Rare Variants and Polymorphisms Associated With C3G and IC-MPGN.

Gene	Mutation/SNP	Function	Classification	References
CFH	Homo-/compound heterozygous SCRs 1–4 (regulatory domain)	Intact surface binding Reduced C3b binding Loss of CFH cofactor and decay accelerating activity	C3G IC-MPGN	(Levy, Halbwachs-Mecarelli et al. 1986, Vogt, Wyatt et al. 1995, Aufl, Schmidt et al. 1997, Dragon-Durey, Fremeaux-Bouchi et al. 2004, Licht, Heiner et al. 2006, Hadley, Mihalovich et al. 2008, Servais, Noel et al. 2012, Bu, Borsa et al. 2016, Iatropoulos, Noris et al. 2016, Iatropoulos, Daina et al. 2018)
CFI	Homozygous Heterozygous	Decreased CFI-mediated C3b degradation	C3G IC-MPGN	(Servais, Noel et al. 2012, Bu, Borsa et al. 2016, Iatropoulos, Noris et al. 2016, Iatropoulos, Daina et al. 2018)
C3	Heterozygous	C3mut-resistant to cleavage by C3bBb C3mut convertase-resistant to CFH-mediated C3 binding with CFI or CFH	C3GN IC-MPGN	(Martinez-Barricarte, Heurich et al. 2010, Bu, Borsa et al. 2016, Iatropoulos, Noris et al. 2016, Iatropoulos, Daina et al. 2018)
CFB	Heterozygous/ Homozygous	Altera C3-FB interaction	C3G IC-MPGN	(Iatropoulos, Noris et al. 2016, Iatropoulos, Daina et al. 2018)
Thrombomodulin	Homozygous	Not tested	DDD	(Iatropoulos, Noris et al. 2016, Iatropoulos, Daina et al. 2018)
DGKE	Homozygous Heterozygous-unclear impact	Not complement mediated	MPGN	(Ozaltin, Li et al. 2013, Bu, Borsa et al. 2016)
At risk SNPs (Reviewed in (Noris and Remuzzi 2015))				
MCP/CD46	Rare SNP	Not tested	C3G IC-MPGN	(Servais, Noel et al. 2012)
CFH	Rare SNP e.g., Y402H (SCR 7)	Not tested	DDD	(Hagemon, Anderson et al. 2005, Abrera-Abeleda, Nishimura et al. 2006, Abrera-Abeleda, Nishimura et al. 2011)
CFHR5	Rare SNP	Not tested	DDD	(Abrera-Abeleda, Nishimura et al. 2006, Abrera-Abeleda, Nishimura et al. 2011, Bu, Borsa et al. 2016)
C3	Rare SNP	Not tested	DDD	(Smith, Alexander et al. 2007, Abrera-Abeleda, Nishimura et al. 2011)
CFHR Fusion Proteins (Reviewed in (Smith, Appel et al. 2018))				
CFHR3-1	CNV CFHR3-1 hybrid gene	Greater degree of CFHR-mediated deregulation	C3GN	(Malik, Lavin et al. 2012, Goicoechea de Jorge, Caesar et al. 2013)
CFHR2-5	CFHR2-5 hybrid gene	Stabilizes C3 convertase, reduced CFH-mediated decay	DDD	(Chen, Wiesener et al. 2014)
CFHR5-CFHR5	CNV Duplication in CFHR5 exons 2/3	Greater degree of CFHR-mediated deregulation	C3G	(Kiele, de Jorge et al. 2010, Goicoechea de Jorge, Caesar et al. 2013)
CFHR1-CFHR1	Internal duplication	Greater degree of CFHR-mediated deregulation	C3G	(Tortajada, Yebenes et al. 2013)
CFHR5-2	CFHR5-2 hybrid gene	Greater degree of CFHR-mediated deregulation	C3GN	(Smith, Appel et al. 2018)
CFHR1-5	CFHR1-2 hybrid gene	Greater degree of CFHR-mediated deregulation	C3G	(Togarsimalemath, Sethi et al. 2017)

CFHR, Complement factor H-related (CFHR) proteins; IC-MPGN, immune complex membranoproliferative glomerulonephritis.

Table_36_5

Table 36.6 Diagnostic Workup for IC-MPGN and C3G

History	Family history of hematuria, glomerulonephritis, and kidney failure Gross hematuria (if yes, concomitant or 2–3 weeks subsequent to infection; number of episodes) Previous urinalysis results? Infection (especially upper respiratory tract) in the preceding 2–3 weeks
Clinical examination	Signs of nephrotic syndrome (peripheral and periorbital edema, weight increase) Blood pressure Presence of frothy urine, reduction in urine output Extrarenal features: partial lipodystrophy, retinal drusen
Laboratory workup	Urinalysis, 24-hour proteinuria, microhematuria Complete blood count, urea, creatinine, protein, IgG, IgA, and IgM Serum C3, C4 Antinuclear antibody, antidouble stranded DNA antibody; serology for hepatitis C and B virus;
Complement workup	C3 nephritic factor (C3Nef), C4Nef and C5Nef if available Anti-FH antibodies; anti-FB antibodies and anti-C3b antibodies if available Soluble C5b9 and total hemolytic complement (CH50); FH, FI, FB levels if available Genetic screening (next-generation sequencing): C3, CFH, CFI, CFB, MCP, CFHR5 in patients with cypriotic origin

Table_36_6